



What is a HbbTV?

Australia has been working on a different broadcast technology called Hybrid Broadcast Broadband TV <http://dgit.in/1pu1mhL>



Athletes testing iWatch

iWatch, supposed to be a health tracker device is being tried out by star athletes at the Cupertino campus <http://dgit.in/TnemuX>

By our powers combined, I'm Captain PSU

Chain two or more power supplies to get the extra juice for your workstation

Mithun Mohandas
readersletters@digit.in

Very often we get to see computer configurations on various websites, forums and magazines. If you'd have paid attention then you'd have noticed that the PSU (Power Supply Unit) or the SMPS (Switching Mode Power Supply) is mentioned accounted for at the end of the configuration chart. This is because one needs to account for the power requirements of each and every component first and then calculate the cumulative power requirements for the entire system before deciding on the power supply. However, there is one thing that almost all of these configurations don't account for and that's the need for extra power in the future.

Imagine that you've built a massively powerful rig and you need all the processing ability that it gives you, a few years down the line you may want to add another graphics card or be a little adventurous and go for three graphics cards. Now to run all of these cards (which may consume about 300W each) you need to buy a really expensive SMPS that can not only power all the cards with ease but also protect them from various electrical anomalies. This new power supply will cost you a bomb and considering that you've just purchased three top of the line graphics cards, adding another ₹20,000 to the bill is the last thing you'd want to do. Or you could simply connect two power supplies together and power

your system easily at a much lower cost. This is also known as "Daisy-chaining" of multiple power supplies and is quite popular with the folks that are into cryptocurrency mining. There are different ways you can go about getting the task accomplished, we'll be covering a DIY approach and also take a look at various after-market solutions available. In order to reduce any problems with this project ensure that the two power supplies are of the same brand and of the same series. This is to eliminate any chances of incompatibility for a wide variety of reasons.

1. Do It Yourself

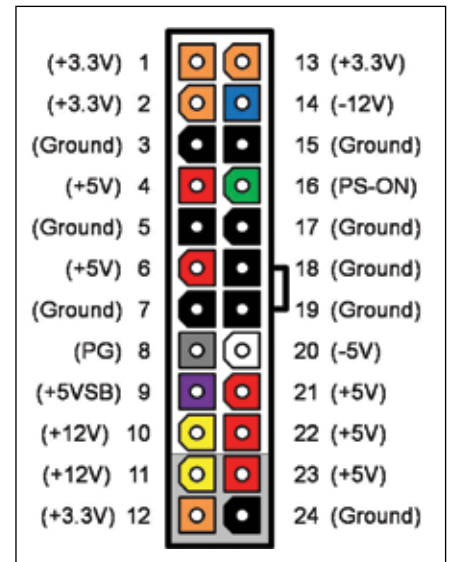
This method requires that you have some experience with basic electronics. You won't have to go out of your way to learn an entire book for this workshop but if it means not blowing up what you're working on then we'd advise you to do a little reading on wiring, electricity loads, parallel and series configurations.

For the purpose of this method you'll need the following.

- 2x Female 24-pin ATX12V power connectors
- 1x Male 24-pin ATX12V power connector
- Wires that are at least 18 AWG for all pins other than pin-11. 22 AWG for pin-11. If the graphics cards consume around 300W then ensure all wires are 16 AWG.



A cable sleeving kit helps with the job



Follow the colour code to avoid mistakes

- ATX connector toolkit
- 2x 10 Ohm 10 Watt resistors

There is a standard for picking the colour for each pin in the ATX standard. Please refer the image above for the same, however, you can choose to ignore and go with whatever colour suits your fancy. Also, AWG (American Wire Gauge) stands for the thickness of the wires, do not compromise on this factor or you'll risk the wires burning out under high load conditions.

In order to link the two power supplies we need to branch out two wires from one power supply and link it to the other. These two pins are pin-16 (green) and pin-17 (black) as shown in the 24-pin ATX12V pinout diagram above. When these two pins are short-circuited the power supply turns on and when they are disconnected the power supply turns off. Ideally, you need to turn on both devices at exactly the same time to prevent current from flowing in the wrong direction.